

Misleading Identifiers Change What Code Traces Write

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Abstract

Language models can write an incorrect intermediate value in a code trace even when the correct value is recoverable from their internal activations. Extending Kudo et al.’s arithmetic analysis to code, we evaluate seven models on matched Python programs that differ only in a variable name. When a variable holds a list’s length but its name suggests a sum, three models write the sum more often than under ordinary names. On these incorrect traces, linear predictors trained on separate programs recover the correct digit with 95.5–99.7% accuracy immediately before the model writes the wrong value. Worked examples that show the operation alongside its result sharply reduce these errors; equally long descriptive annotations leave large effects. These findings distinguish values recoverable inside a model from those reported in its trace, and show that demonstrating the operation helps the written trace follow the program despite a misleading name.

1 Introduction

A variable named `sum_all` is assigned `len(xs)`. For `[5, 3]`, its value is 2, regardless of its name. Suppose a language model is asked to trace the program: write each assignment’s value, then the return value. In Figure 1, OLMo-2-7B writes 8 for this variable, following the sum suggested by the name. We call this a *name error*.

Changing only the worked examples changes this trace. When examples show values alone, the model writes 8. When they also show the operation used to obtain each value, it writes 2. The program, input and model are unchanged. This contrast motivates our study: how do variable names and the format of worked examples affect which value a model writes?

Earlier work shows that misleading identifiers reduce code accuracy (Le et al., 2025; Lam et al., 2025; Le et al., 2026). The wrong trace raises a

further question: when the model writes 8, can we still recover 2 from its internal activations? We build on Kudo et al. (2026)’s analysis of intermediate values during arithmetic reasoning. Their probes test what values can be recovered from activations; their patching experiments replace activations to test how outputs change. We apply these tools to code with a deliberate conflict between a variable’s assigned operation and its name.

We train a *linear probe*, a small predictor of the true digit, on separate programs. It reads activations immediately before the model writes a value. We show that this predictor usually returns the computed digit even when the trace follows the misleading name. Showing the operation in worked examples sharply reduces those wrong writes; extra descriptive text of the same length leaves large effects. The worked examples therefore change whether the trace writes the value computed by the code or the value suggested by the name.

2 Study design

Each problem is a short Python function over a list of distinct digits. Assignments apply list aggregates (`len`, `sum`, `max`, `min`) or integer operations, followed by a return and a call. True intermediate and final values are single digits. The main level has three dependent assignments; Appendix A gives the full design. We test seven instruction-tuned models with 1,000 matched problems per level and target, with three fixed sets of three worked examples. CodeLlama-34B supplies an additional scale check.

Each matched pair differs only in one target identifier. A congruent name implies the assigned operation; a misleading name implies another. Its name-suggested value must differ from every value, constant and input element in the problem. This prevents copying from explaining

(a) conflicting values

```
def f(xs):  
    sum_all = len(xs)  
    ww = sum_all + 3  
    qq = ww + 4  
    return qq  
f([5, 3])
```

code: $\text{len}(xs) = 2$

name: $\text{sum}(xs) = 8$

the call returns 9

(b) OLMo-2-7B's traces

Examples show: $v = 3$

$\text{sum_all} = 8, \text{ww} = 11, \text{qq} = 15$

answers 15 (wrong)

Examples show: $v = \text{len}(xs) = 3$

$\text{sum_all} = \text{len}(xs) = 2, \text{ww} =$

$\text{sum_all} + 3 = 5, \text{qq} = \text{ww} + 4 = 9$

answers 9 (correct)

Same problem and worked examples.
Only the examples' format changes.

Figure 1: (a) Code and name imply different values. (b) OLMo-2-7B's traces under demonstrations showing values alone (red) or expressions with values (green).

a name error and excludes max and min as name-suggested operations on the aggregate step. We also check which operation each model associates with each identifier. The original main names fail the planned 80% panel gate, a selection-rule deviation. We disclose it and independently replicate with screened names and fresh programs (Appendix L).

In the *direct* regime, models write the return value. In *trace*, they first write each assignment's value, as in $\text{qq} = 3, v = 6, vv = 9$. The *trace-expr* comparison adds the expression to each demonstrated step, as in $\text{qq} = \text{len}(xs) = 3$. REPL, comment and free-prose formats test the effect's scope (Appendix D).

Our main measure is *excess name errors*: the rate at which a misleading trace writes the name-suggested value at the target step, minus the neutral twin's rate of writing that same digit. We measure name errors similarly in the direct regime. We call an effect reliable when its sign agrees across all three demonstration sets and a 95% cluster-bootstrap interval excludes zero. To support a small-effect claim, a 90% interval must fit inside the preregistered ± 2 -point margin. Other cases remain inconclusive (Appendix G).

For three models, a linear probe predicts the target digit from a hidden state at one token and layer. Training uses disjoint computations and

three probe seeds; layers are selected on neutral programs in other folds before each held-out computation is evaluated. We read the *decision token*: the = immediately before the first traced value. The generated and cached token prefixes must agree through that token; no value digit is included. Patching replaces the name tokens or this position with a matched neutral twin's activations. A neutral-for-neutral patch measures disruption. Controls and implementation details are in Appendix E.

3 Results

3.1 Terse traces follow a misleading name

Table 1 shows the main level. Under sum-like names, length-computed variables produce +58.4, +43.4 and +12.4 points of excess name errors in OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B. On neutral programs, their trace accuracies are 95.7, 97.4 and 99.6%, respectively. These models usually trace neutral programs correctly.

The sum is not these models' usual error on a length step. On wrong neutral traces, sum writes account for only 14–32% of errors; maximum values and off-by-one lengths are more common. The sum-like name changes which wrong value they write. CodeLlama-7B's larger effect occurs on a maximum-computed variable (+10.7 points);

Model	Trace acc.	Direct: errors	Trace: len→sum	Trace: max→sum
OLMo-2-7B	95.7	+0.4	+58.4* [+53.0, +63.5]	+1.9 [+1.1, +3.0]
Llama-3.2-3B	97.4	+0.8	+43.4* [+38.5, +48.1]	+0.0 [+0.0, +0.0]
Llama-3.1-8B	99.6	+0.5	+12.4* [+8.8, +16.0]	+0.0 [+0.0, +0.0]
CodeLlama-7B	91.7	+1.0	+3.7* [+2.2, +5.4]	+10.7* [+7.9, +13.7]
CodeGemma-7B	99.2	+1.5*	+0.1 [+0.0, +0.4]	+0.0 [+0.0, +0.0]
Gemma-3-4B	99.1	+0.8	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
OLMo-2-1B	37.8	+3.1*	+3.7* [+2.2, +5.5]	+4.2* [+2.6, +5.9]
CodeLlama-34B	98.9	–	+0.0 [+0.0, +0.0]	+1.8* [+0.7, +3.1]

Table 1: Level 5 means across three demonstration sets. *Trace acc.*: neutral accuracy; other columns: excess name errors (percentage points [95% intervals]). *: consistent signs and intervals excluding zero. Trace columns pair length or maximum with sum-like names.

CodeGemma-7B shows that pairing at level 3. The full pairing matrix is in Appendix C. Direct-answer name-error effects are much smaller: -0.2 to +3.1 points at this level.

3.2 The correct digit is decodable on error cases

On generations that write the name-suggested value, neutral-trained probes recover the computed digit with 96.4%, 95.5% and 99.7% accuracy in OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B. The prefixes match through the decision token in all 2,565 generations across the three demonstration sets. The cached state therefore also precedes the model’s actual value write. Appendix K reports clustered intervals and the correct-write and other-error strata.

Five-fold layer selection excludes each test computation. Neutral name-identity controls score 1.00, 1.00 and 0.88. Recovering the correct digit on an error establishes decodability, not its use in generation.

Patching independently tests sensitivity to the name and the writing position (Figure 2). Replacing name tokens at layer 0 removes 97% and 100% of name errors in the two models with aligned twins. Name-site removal declines in later layers. At the decision token, removal is weak early and strong from the middle of the network onward; all-layer replacement removes 97–100%. The neutral-word control changes at most 1.0% of outputs. The patches identify positions sensitive to the misleading name. They replace an entire hidden-state vector, so they cannot identify which feature changes the output.

3.3 Showing the expression reduces the effect

We keep the programs, models and demonstration sets fixed and change the examples from $v = 3$

Model	trace	REPL	comment	trace-expr
OLMo-2-7B	+58.4*	+9.5*	+0.0	+3.4*
Llama-3.2-3B	+43.4*	+12.6*	+6.8*	+0.4
Llama-3.1-8B	+12.4*	+0.2	+0.1	+0.0

Table 2: Length-to-sum excess name errors (percentage points) by demonstration format; programs and examples fixed. Intervals: Table 6.

to $v = \text{len}(xs) = 3$. Name-error rates fall by 55, 43 and 12 points in the three models (Table 2). OLMo-2-7B retains +3.4 points of excess; the two Llama effects approach zero. Length-matched neutral annotations leave 56.6 and 49.6 points of excess in OLMo and Llama-3.2. Valid numeric elaborations ($v = 3 + 0 = 3$) lower it to 41.3 and 14.9 points, still above source-expression demonstrations (Appendix J). Extra tokens do not reproduce the expression benefit; adding arithmetic accounts for part of it.

With 3 independently screened names on 200 fresh computations, trace excess in OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B is 91.1, 61.5 and 16.7 points; expression excess is 0.8, 0.0 and 0.0. CodeGemma shows no observed excess in either format (Appendix L).

REPL transcripts also omit expressions, yet their effects are several times smaller; comment traces reduce them further. The larger checkpoint has a smaller maximum-step effect: CodeLlama-34B has +0.0 excess on length steps and +1.8 [+0.7, +3.1] on maximum steps. The latter interval crosses the equivalence margin. Renamed CRUXEval functions show no detectable accuracy effect in direct-answer or prose regimes; those tests do not include the synthetic trace format (Appendix F).

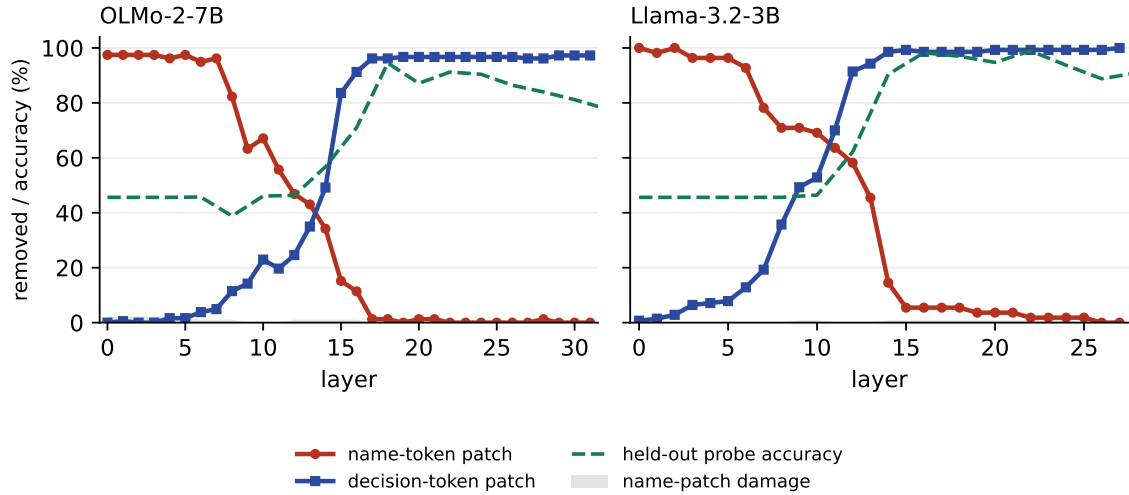


Figure 2: Name-error removal at name tokens (red) and the decision token (blue). Dashed green: probe accuracy on held-out misleading-name programs; grey: damage to correct answers. Llama-3.1-8B’s `sum_all` lacks an aligned twin for name-site patching.

4 Related work

Misleading identifiers affect real-code accuracy (Le et al., 2025; Lam et al., 2025; Le et al., 2026; Orvalho and Kwiatkowska, 2025; Gao et al., 2023). Our synthetic tasks separate computed and name-suggested values. Kudo et al. (2026) study arithmetic readouts; Anonymous (2026) find their association with arithmetic name errors sensitive to task competence. Neither establishes a mechanism shared with code traces. Faithfulness studies perturb prompts and chains (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025); we intervene at the value-writing position. Stroop studies focus on next-token prediction (Stroop, 1935; Geirhos et al., 2019; Wang, 2026; Hu et al., 2026).

5 Conclusion

Terse code traces can write a name-implied value while the computed digit remains decodable. Patches and expression demonstrations reduce these errors; the format contrast replicates with independently screened names. Length-matched annotations leave large effects, while numeric elaboration helps partly. The finding concerns the tested names and formats; it does not establish a general failure of code computation.

Limitations

- **Scope.** Programs are synthetic with single-digit values; the largest effect uses a constructed trace format. REPL and com-

ments show smaller effects. The 89-function CRUXEval check bounds direct-answer and prose accuracy, not synthetic trace errors. Models reach 34B; the panel excludes Chinese-origin models, including strong code models in Le et al. (2026).

- **Internal measurements.** Probes on three smaller models establish decodability, not use. Error readouts select layers independently; full-neutral curves and name controls are descriptive. Patches replace mixed representations and may have side effects. Llama-3.1-8B’s `sum_all` lacks an aligned twin for name-site patching.
- **Design deviations.** The registered REPL prediction failed. All three main-cell names fail the planned panel meaning gate, leaving its filtered estimate unavailable. Screened-name replication supports the format contrast but does not repair that deviation; probes and patches retain the original names (Appendix L).

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Level	Neutral program (body)	Traced steps
1	<code>v = len(xs); return v</code>	1
2	<code>v = len(xs); w = v * 2; return w</code>	2
3	<code>v = len(xs); w = max(xs); u = v + w; return u</code>	3
4	level 3 with an unused fourth assignment	3
5	<code>v = len(xs); w = v * 2; u = w + 3; return u</code>	3
6	level 5 with the misleading name on the middle step	3

Table 3: The levels, one example each. The aggregate in the first statement is any of len, sum, max, min; unary operations are $\ast$, $//$, $+$, $-$. The misleading name sits on the first statement at levels 1–5 and on the second at level 6.

Family	Names	Implied operation
len	count, n_items, length	<code>len(xs)</code>
sum	total, sum_all, acc	<code>sum(xs)</code>
max	largest, max_val, peak	<code>max(xs)</code>
min	smallest, min_val, low	<code>min(xs)</code>
double	double, twice	<code>2 * a</code>
half	half	<code>a // 2</code>
succ	succ, next_val	<code>a + 1</code>
pred	pred, prev_val	<code>a - 1</code>
add	combined, both	<code>a + b</code>
sub	diff, gap	<code>a - b</code>
mul	product, scaled	<code>a * b</code>

Table 4: Identifier families and their suggested operations. The gate (Appendix B) scores each name by the log-probability each model assigns to the candidate right-hand sides; 12 of 25 names are read as intended by at least 80% of models. The min and max names cannot produce name errors on the aggregate step (their value is an input element) and are used as congruent and alternative-name names.

C Operation and name pairings

Table 5 reports all tested aggregate/name pairings. The reverse of the main length-to-sum conflict is weak at level 5: sum-computed variables with length-like names have excess within two points for models above the neutral-accuracy floor. At level 3, OLMo-2-7B has +4.4 points of excess in that pairing. Gemma-3-4B shows no main-cell effect, consistent with its identifier-meaning check. OLMo-2-1B’s neutral accuracy is only 37.8%, making its small, broadly distributed effects harder to interpret as a specific trace failure.

D Demonstration formats

All few-shot comparisons hold programs, demonstration identities and model weights fixed. The trace writes `qq = 3`; trace-expr writes `qq = len(xs) = 3`; REPL writes `>>> qq` followed by 3; comment mode annotates each assignment with its value. Prose asks for step-by-step reasoning without demonstrations. Table 6 gives the paired excess and interval for each few-shot format.

On the main conflict, REPL excess is +9.5 and +12.6 points for OLMo-2-7B and Llama-3.2-3B, compared with +0.2 for Llama-3.1-8B. Comments

leave +6.8 points in Llama-3.2-3B and near-zero excess in the other two. Prose has little initial name errors, so asking it to state or recompute values does not test removal of the large trace effect. The prediction that REPL would stay within a factor of two of trace was not supported; the comment prediction held only for two models.

The paired trace-minus-trace-expr name-error-rate reductions are 55 points [49.9, 60.2], 43 [38.2, 47.7] and 12 [8.8, 16.0] for OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B. These 95% intervals keep each matched program’s demonstration-seed repeats together.

E Probe and patching details

We follow Kudo et al.’s 10-class linear-probe recipe, with three seeds and neutral-only training. Each probe predicts a target digit from one residual-stream position and layer. Name-identity controls use deterministic random labels keyed on the identifier (Hewitt and Liang, 2019). Neutral names and values vary independently, and misleading names are absent from training. The original implementation reused test programs in training. The reported reruns replace those scores: they train on 2,000 independent neutral programs

Model	len→sum	max→sum	min→sum	sum→len	max→len
OLMo-2-7B	+17.6* / +58.4*	+0.9 / +1.9	+0.0 / +0.2	+4.4* / +0.0	+0.0 / +0.0
Llama-3.2-3B	+23.6* / +43.4*	+1.1 / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
Llama-3.1-8B	+1.5 / +12.4*	+0.7 / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
CodeLlama-7B	+0.9 / +3.7*	+7.5 / +10.7*	+1.0 / +1.7	+1.5 / +0.0	+0.9 / +0.6
CodeGemma-7B	+0.0 / +0.1	+10.8* / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
Gemma-3-4B	+0.0 / +0.0	+0.2 / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
OLMo-2-1B	+1.2 / +3.7*	+2.8* / +4.2*	+0.3 / -0.2	+10.6* / +11.1*	+9.6* / +5.6*

Table 5: Excess name errors at the aggregate step, trace regime, level 3 / level 5, three sets of worked examples pooled; * as in Table 1.

Model	trace	REPL	comment	trace-expr
OLMo-2-7B	+58.4* [+53.0, +63.5]	+9.5* [+6.9, +12.0]	+0.0 [+0.0, +0.0]	+3.4* [+2.0, +5.1]
Llama-3.2-3B	+43.4* [+38.5, +48.1]	+12.6* [+9.4, +15.9]	+6.8* [+4.4, +9.2]	+0.4 [+0.0, +0.9]
Llama-3.1-8B	+12.4* [+8.8, +16.0]	+0.2 [+0.0, +0.6]	+0.1 [+0.0, +0.4]	+0.0 [+0.0, +0.0]

Table 6: Table 2 with 95% bootstrap intervals over matched problems.

and evaluate 285 misleading-name programs, with overlap guards that ignore name spelling. Table 7 reports the name controls alongside digit accuracy.

The main scores use demonstration set 7. Repeating training and evaluation with sets 11 and 13 gives true-digit accuracies of 95–97%, 94–100% and 100–100% across the three sets. Each repeat uses the same independent training programs and three probe seeds; its layer is selected by that set’s neutral accuracy. These repeats assess prompt sensitivity, not a causal role for the decoded digit. Table 7 gives both conditions’ readouts and controls.

We supply the correct trace token by token when caching probe inputs. The decision position is the = before the first traced value; later values cannot affect this position under causal attention. Patching uses only the prompt and this trace prefix. Name patches replace every aligned occurrence of the identifier; decision patches replace only that writing position. Both use one layer at a time and include a neutral-for-neutral disruption control (Zhang and Nanda, 2024). The alternative-name control is uninformative in the main cell: every sum-family identifier implies the same sum, so there is no distinct source name-suggested value. Llama-3.1-8B’s affected two-token name lacks aligned neutral twins for the name-site intervention.

F Scope checks

Operation and scale. Moving the misleading name to a unary middle step produces no reliable effect for models above the accuracy floor; the largest excess is +0.6 points. CodeLlama-34B’s

length-to-sum effect is +0.0, while its maximum-to-sum effect is +1.8 points [+0.7, +3.1]. The latter is a reliable positive effect, but its interval extends beyond the two-point margin. Its small point estimate cannot be reported as equivalence. The larger-checkpoint result supports the registered maximum-to-sum direction and reliable-effect criterion, while its magnitude is much smaller than CodeLlama-7B’s.

Natural code. We select 89 CRUXEval functions with length, loop-counter or maximum assignments, rename the target by AST rewriting to v or total, and verify execution after renaming. Direct and prose accuracy contrasts all have intervals covering zero on the 84 functions whose renamed target is used downstream. Their endpoints stay within 10 and 13 points, respectively (Table 9). These are bounded results for those functions and prompts; they do not test the terse few-shot trace used in the synthetic study.

G Effect and equivalence checks

Each contrast subtracts the neutral twin’s rate for the same digit. We resample matched problems with demonstration-seed replicates together and use 95% cluster-bootstrap intervals. Reliability also requires a consistent sign across all three demonstration sets.

Equivalence applies the ± 2 -point margin to 90% intervals from the same cluster bootstrap (two one-sided 5% tests), using 2,000 draws. Across 192 tested contrasts, 150 intervals lie wholly inside the margin, 36 contrasts have reliable effects without meeting that bound, and 6

Model	Set	Layer	Neutral	Control	Misleading	Control	Majority
OLMo-2-7B	7	18	99.5	100.0	94.5	33.0	45.6
OLMo-2-7B	11	18	97.3	100.0	95.4	0.0	45.6
OLMo-2-7B	13	18	99.8	100.0	97.4	23.4	45.6
Llama-3.2-3B	7	24	98.8	99.7	93.7	20.0	45.6
Llama-3.2-3B	11	20	100.0	99.8	95.4	5.0	45.6
Llama-3.2-3B	13	28	99.4	100.0	99.9	1.9	45.6
Llama-3.1-8B	7	16	100.0	87.8	99.9	33.0	45.6
Llama-3.1-8B	11	20	100.0	81.6	100.0	0.1	45.6
Llama-3.1-8B	13	20	100.0	87.2	100.0	3.0	45.6

Table 7: Disjoint code probes, three demonstration sets. Neutral and misleading are true-digit accuracy; each control column gives name-identity accuracy for that condition. Majority predicts the most frequent digit in training on the misleading test programs. Rates are percentages, averaged over three probe seeds. Layers are selected by neutral accuracy.

Model	Name tokens			Decision token			
	L0	damage	<10% by	50% at	90% at	all layers	damage
OLMo-2-7B	97%	2%	L17	L15	L16	97%	2%
Llama-3.2-3B	100%	0%	L15	L10	L12	100%	0%
Llama-3.1-8B	—	—	—	L8	L9	100%	0%

Table 8: Layers behind Figure 2. Name tokens: name errors removed by replacing every token of the name at layer 0, damage to correct instances, and the first layer after which removal falls below 10%. Decision token: first layer at which a single-layer replacement removes 50% and 90% of name errors, removal with every layer replaced, and its damage. Llama-3.1-8B’s name errors fall on a two-token name without a token-aligned twin.

meet neither criterion. A small estimate or an interval covering zero does not by itself establish equivalence.

H Generated-prefix audit and initial readouts

For 2565 of 2565 misleading-name generations, the token prefix through the marker before the first value exactly matches the cached trace. This verifies that the cached state at that marker also precedes the generated value. No target value token is included. These initial readouts select layers on the full neutral evaluation set; the main text instead uses independent cross-fitted selection (Appendix K). We split the generated writes into name-suggested values, correct values and other errors. Table 10 averages three probe seeds and pools demonstration sets 7, 11 and 13. Intervals resample matched programs with all their demonstration repeats (4,000 draws). These conditional readouts establish available information on observed errors; they do not establish its causal use.

I Sensitivity to identifier meanings

None of the three main-cell names (total, sum_all, acc) meets the originally planned 80%

panel gate. Its filtered estimate is therefore unavailable, rather than zero. Table 11 retains the full-data estimate and reports each name and each model’s own meaning gate. Gates use the original saved meaning checks. Intervals cluster matched programs across the three demonstration sets. This exploratory sensitivity does not replace the planned panel gate or remove the deviation from it.

J Length and numeric elaboration controls

We add neutral annotations to each demonstrated value, matching the expression demonstrations’ token counts, and separately add valid numeric elaborations of the form name = value + 0 = value. The test program and Trace: prefix are unchanged. Greedy decoding uses the same 285 misleading programs and neutral twins for each format. Table 12 pools three demonstration sets; changes relative to the original trace and expression trace are paired on the same programs. The length-matched annotation controls retain much larger excess than expression demonstrations in both models. Numeric elaboration reduces excess relative to the original trace but leaves a larger effect than source expressions. Token count alone

Model	Direct acc.	Direct Δ	Prose acc.	Prose Δ
OLMo-2-7B	10	+1.2 [+0.0, +3.6]	22	-1.2 [-9.5, +8.3]
Llama-3.2-3B	19	+1.2 [-2.4, +4.8]	25	+3.6 [-3.6, +10.7]
Llama-3.1-8B	22	-3.6 [-9.5, +2.4]	40	+0.0 [-9.5, +9.5]
CodeLlama-7B	18	+2.4 [+0.0, +6.0]	13	+2.4 [-4.8, +9.5]
CodeGemma-7B	18	+0.0 [-4.8, +4.8]	22	+2.4 [-8.3, +13.1]
Gemma-3-4B	15	-2.4 [-6.0, +0.0]	45	+4.8 [-1.2, +11.9]
OLMo-2-1B	16	-3.6 [-8.3, +0.0]	3	+6.0 [+0.0, +11.9]

Table 9: 89 CRUXEval functions that assign a variable from `len( . . . )` (48), a loop counter (38) or `max( . . . )` (3), of which 84 use it downstream; each rendered with the original name, the neutral name `v`, and the misleading name `total`. Accuracy on all neutral renderings, and the paired misleading-minus-neutral difference in points on the downstream-used subset, with a bootstrap interval over functions.

Model	Generated write	Rows	Programs	True-digit accuracy (%)
OLMo-2-7B	name error	503	193	96.4 [94.8, 97.8]
OLMo-2-7B	correct write	320	135	96.7 [94.4, 98.5]
OLMo-2-7B	other error	32	24	77.1 [61.7, 90.5]
Llama-3.2-3B	name error	372	169	95.7 [93.8, 97.4]
Llama-3.2-3B	correct write	440	204	98.3 [97.0, 99.3]
Llama-3.2-3B	other error	43	29	82.2 [71.2, 91.7]
Llama-3.1-8B	name error	106	45	99.7 [99.0, 100.0]
Llama-3.1-8B	correct write	749	261	100.0 [100.0, 100.0]
Llama-3.1-8B	other error	0	0	–

Table 10: Disjoint probes at the verified generated prefix. A dash denotes an empty stratum. The denominator is observed writes within each stratum, not independent demonstration repeats.

does not reproduce the expression benefit, and adding valid arithmetic accounts for part of it; these two contrasts do not isolate a single mechanism.

K Independent probe layer selection

The main-text error readouts use five-fold cross-fitting. A fixed SHA256 hash assigns canonical program computations to folds, keeping renamed versions and demonstration repeats together. For each held-out fold, we select the layer with the highest neutral accuracy on the other four folds, averaging three probe seeds and breaking ties toward the lowest layer. Neither the held-out neutral labels nor its misleading writes enter that choice. Probe training remains disjoint from every evaluation computation. Each program is scored only with its held-out fold’s layer. The generated-prefix audit from Appendix H still applies. Table 13 reports the conditional write strata and paired changes from the earlier full-neutral selection. Intervals use 4,000 canonical-program bootstrap draws, keeping demonstration repeats together. They condition on the fixed fold selectors and do not refit layer selection inside each bootstrap. Name controls and the full layer curves remain descriptive measurements in Appendix E.

L Independent replication with screened names

We fix 16 fresh sum-family candidates before scoring them with the original candidate-continuation meaning check (Appendix B). The panel is the same seven models used for that gate. A name qualifies when `sum` is the highest-scoring continuation in at least 80% of the panel; ties follow the original scoring order. Two additional function-name contexts and neutral names are diagnostic and do not affect selection. Of the candidates, 14 qualify. We freeze the first 3 qualifying names in the fixed order: `sum_of_values`, `sum_of_items`, `sum_of_elements`. No trace outcomes enter selection. The replication uses 200 new length-computed programs. Their canonical computations are unique and absent from the original level-5 training and test corpus and all three demonstration sets. True and name-suggested values are checked by execution and retain the original value-exclusion rules. Every selected name is crossed with every program; its neutral twin differs only in the target identifier. The three affected models and the prespecified CodeGemma-7B control use the same programs and demonstration sets 7, 11 and 13 under trace and expression demonstrations. Table 15 pools names and demonstration

Model	All	Model gate	total	sum_all	acc
Gemma-3-4B	0.0 [0.0, 0.0]	–	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
CodeGemma-7B	0.1 [0.0, 0.4]	0.1 [0.0, 0.4]	0.0 [0.0, 0.0]	0.3 [0.0, 1.0]	0.0 [0.0, 0.0]
Llama-3.1-8B	12.4 [8.9, 16.0]	12.4 [8.9, 16.0]	0.0 [0.0, 0.0]	34.0 [26.0, 42.3]	0.0 [0.0, 0.0]
OLMo-2-1B	3.7 [2.2, 5.5]	5.0 [1.8, 8.9]	5.0 [1.8, 8.9]	5.8 [2.6, 9.3]	0.0 [0.0, 0.0]
OLMo-2-7B	58.4 [53.1, 63.4]	59.3 [52.7, 65.8]	56.4 [47.9, 64.9]	96.2 [93.9, 98.1]	15.3 [9.6, 21.8]
CodeLlama-7B	3.7 [2.1, 5.5]	3.7 [2.1, 5.5]	1.8 [0.4, 3.2]	8.7 [4.5, 13.1]	0.0 [0.0, 0.0]
Llama-3.2-3B	43.4 [38.5, 48.0]	43.4 [38.5, 48.0]	30.1 [23.8, 36.9]	76.3 [70.2, 82.4]	18.4 [11.9, 25.7]

Table 11: Excess name errors in percentage points [95% interval], level 5 length-to-sum. Empty subsets are shown as dashes. The saved JSON also reports each demonstration set and the all-three-sets sign rule.

Model	Format	Excess	Change vs trace	Change vs expression
OLMo-2-7B	neutral annotation	56.6 [51.3, 61.6]	-1.8 [-3.4, -0.1]	53.2 [48.1, 58.2]
OLMo-2-7B	numeric elaboration	41.3 [36.4, 46.0]	-17.1 [-21.1, -12.9]	37.9 [33.2, 42.5]
Llama-3.2-3B	neutral annotation	49.6 [44.7, 54.3]	6.2 [4.1, 8.4]	49.2 [44.3, 53.9]
Llama-3.2-3B	numeric elaboration	14.9 [12.0, 17.8]	-28.5 [-32.7, -24.3]	14.5 [11.7, 17.4]

Table 12: Excess name errors and paired changes in percentage points [95% interval]. Bootstrap clusters keep all demonstration repeats for each matched program together. These controls assess length and numeric elaboration; they do not by themselves establish how expressions change generation.

sets, clustering all repeats of each computation in 4,000 bootstrap draws. Table 16 reports each identifier. The format contrast replicates on independently selected names, but names and programs differ from the original experiment, so differences in effect magnitude across the two samples are descriptive. This exploratory replication does not retroactively repair the original selection-rule deviation. Probes and patches concern the original identifiers, not these new names.

Model	Write	Rows	Computations	Accuracy (%)	Change (points)
OLMo-2-7B	name error	503	193	96.4 [94.9, 97.9]	0.0 [0.0, 0.0]
OLMo-2-7B	correct write	320	135	96.7 [94.4, 98.4]	0.0 [0.0, 0.0]
OLMo-2-7B	other error	32	24	77.1 [61.6, 90.6]	0.0 [0.0, 0.0]
Llama-3.2-3B	name error	372	169	95.5 [93.6, 97.3]	-0.2 [-0.9, 0.4]
Llama-3.2-3B	correct write	440	204	98.0 [96.6, 99.1]	-0.3 [-0.8, 0.0]
Llama-3.2-3B	other error	43	29	82.2 [71.1, 91.7]	0.0 [0.0, 0.0]
Llama-3.1-8B	name error	106	45	99.7 [99.0, 100.0]	0.0 [0.0, 0.0]
Llama-3.1-8B	correct write	749	261	100.0 [99.9, 100.0]	-0.0 [-0.1, 0.0]
Llama-3.1-8B	other error	0	0	–	–

Table 13: True-digit readouts with independent layer selection. Changes compare the same observed writes with the earlier full-neutral layer choice. A dash denotes an empty stratum. These readouts establish decodability, not causal use.

Candidate	Sum agreement	Qualifies	Selected
sum_of_values	6/7	yes	yes
sum_of_items	7/7	yes	yes
sum_of_elements	7/7	yes	yes
sum_of_xs	7/7	yes	no
sum_of_list	6/7	yes	no
list_sum	6/7	yes	no
input_sum	6/7	yes	no
items_sum	6/7	yes	no
elements_sum	5/7	no	no
sum_values	5/7	no	no
sum_result	6/7	yes	no
sum_total	6/7	yes	no
total_sum	6/7	yes	no
xs_sum	7/7	yes	no
sum_xs	7/7	yes	no
summation	6/7	yes	no

Table 14: The entire fixed candidate list, in selection order. Qualification uses the original canonical meaning prompt. Selection retains the first three qualifying candidates without consulting behavior.

Model	Trace excess	Expression excess	Expression minus trace
OLMo-2-7B	91.1 [88.4, 93.6]	0.8 [0.3, 1.3]	-90.3 [-92.9, -87.6]
Llama-3.2-3B	61.5 [56.1, 66.9]	0.0 [0.0, 0.0]	-61.5 [-66.9, -56.1]
Llama-3.1-8B	16.7 [12.8, 20.8]	0.0 [0.0, 0.0]	-16.7 [-20.8, -12.8]
CodeGemma-7B	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Table 15: Independent screened-name replication: excess name errors and paired format changes in percentage points [95% interval]. All selected identifiers and three demonstration sets are pooled over 200 fresh computations. A zero bootstrap interval records no observed paired name errors; it is not a population-level absence or equivalence claim.

Model	Identifier	Trace excess	Expression excess
OLMo-2-7B	sum_of_values	91.7 [88.8, 94.3]	0.2 [0.0, 0.5]
OLMo-2-7B	sum_of_items	89.2 [86.0, 92.0]	2.0 [0.8, 3.3]
OLMo-2-7B	sum_of_elements	92.5 [89.8, 94.8]	0.2 [0.0, 0.5]
Llama-3.2-3B	sum_of_values	61.7 [56.0, 67.3]	0.0 [0.0, 0.0]
Llama-3.2-3B	sum_of_items	55.5 [50.0, 61.3]	0.0 [0.0, 0.0]
Llama-3.2-3B	sum_of_elements	67.3 [61.7, 73.0]	0.0 [0.0, 0.0]
Llama-3.1-8B	sum_of_values	12.8 [9.3, 16.7]	0.0 [0.0, 0.0]
Llama-3.1-8B	sum_of_items	22.8 [18.0, 27.7]	0.0 [0.0, 0.0]
Llama-3.1-8B	sum_of_elements	14.5 [10.5, 18.7]	0.0 [0.0, 0.0]
CodeGemma-7B	sum_of_values	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
CodeGemma-7B	sum_of_items	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
CodeGemma-7B	sum_of_elements	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Table 16: Per-identifier replication, pooling the same three demonstration sets. Effects are percentage points [95% canonical-program bootstrap interval]. All affected-model trace means are positive in every demonstration set; per-set estimates are retained in the saved summary.